Unit: 6 Unit 8: Quadratics
Topic: Solving by Quadratic Formula (Discriminant)

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Solve $x^2 + 5x + 6 = 0$ using the quadratic formula.
(A) $x = -2, x = -3$
(B) $x = -1, x = -6$
(C) $x = 2, x = 3$
(D) $x = 1, x = 6$
(E) $x = -2, x = 6$

Q2. What is the discriminant of $x^2 - 4x + 4 = 0$?
(A) 0
(B) 4
(C) 8
(D) 16
(E) 20

Q3. Solve $x^2 + 2x + 1 = 0$ using the quadratic formula.
(A) $x = -1$
(B) $x = 1$
(C) $x = -1, x = 1$
(D) $x = 0$
(E) $x = 2$

Q4. What type of solutions does $x^2 + 1 = 0$ have?
(A) Two real solutions
(B) One real solution
(C) No real solutions
(D) Infinite solutions
(E) Two complex solutions

Q5. What is the value of $b^2 - 4ac$ in $x^2 - 3x - 4 = 0$?
(A) 1
(B) 9
(C) 17
(D) 25
(E) 33

Q6. Solve $x^2 - 7x + 12 = 0$ using the quadratic formula.
(A) $x = 3, x = 4$
(B) $x = -3, x = -4$
(C) $x = 1, x = 12$
(D) $x = -1, x = -12$
(E) $x = 2, x = 6$

Q7. What is the discriminant of $x^2 + 2x + 5 = 0$?
(A) -16
(B) -4
(C) 4
(D) 16
(E) 20

Q8. How many solutions does $x^2 - 4 = 0$ have?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 0

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 6x + 8 = 0$ using the quadratic formula and interpret the discriminant.

Q10. Find the discriminant and determine the number and type of solutions for $x^2 - 2x - 6 = 0$.

Chef's Tips

- Ensure students have a calculator for complex calculations.
- Emphasize the importance of checking the discriminant before solving.
- Remind students to simplify their answers.